

Income-Related Performance Gaps in Vision-Language Models

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Abstract

We test five pretrained vision-language model families on a category-balanced, 168-image Dollar Street benchmark spanning four income quartiles. Overall classifier accuracy was 86.3–86.9%, yet Q4-minus-Q1 gaps differed by model: CLIP 16.7 percentage points (95% CI 2.4 to 28.6), Qwen 2.4 points (-7.1 to 14.3), and InternVL 0.0 points (-9.5 to 9.5). Caption term recall favored Q4 by 21.4–52.4 points. Blinded human review found higher BLIP semantic correctness than automatic recall, while the tested prompt produced no reliable improvement. Results are model-, task-, and metric-specific and do not establish a causal income effect.

1. Introduction and Related Work

Household objects vary across homes: a stove can be built in, tabletop, or solid-fuel, and similar variation occurs in roofs, lights, switches, footwear, and waste containers. Dollar Street was designed to expose geographic and socioeconomic visual diversity [6]. Prior work reported income-related differences in CLIP alignment and retrieval [4]; our study fixes equal category counts, compares classification, captioning, and open-vocabulary detection on the same images, and adds human review.

CLIP provides transferable zero-shot recognition through contrastive image-text pretraining [5]. BLIP combines visual understanding and language generation [3], enabling our paired caption-prompt intervention. Qwen2.5-VL [1] and InternVL3.5 [7] add instruction-following multimodal generation, while YOLO-World [2] reframes detection as open-vocabulary matching. Together, these architectures test whether one income pattern persists across output types and metrics.

RQ1: How do task metrics vary from the lowest (Q1) to highest (Q4) income quartile? **RQ2:** Does prefixing BLIP with “the main household object in this image is” improve correctness overall and in Q1 without increasing hallucination? Income only groups images and is not treated as causal.

2. Method

Data. From the public 1,600-row Dollar Street test split, we removed 45 multi-label rows and formed quartiles from 1,555 eligible records at PPP-adjusted monthly-income cut points 210.67, 685, and 1,841. The locked manifest contains 168 unique images from 44 countries and six categories (roof, light source, stove, trash container, switch, footwear), with seven images in every category-quartile cell. Source rows and a checksum are retained.

Models. Without fine-tuning, CLIP *openai/clip-vit-base-patch32* performed six-way zero-shot classification with “a photo of a {label}.” BLIP *Salesforce/blip-image-captioning-base* generated baseline and prompted captions. Qwen *Qwen/Qwen2.5-VL-3B-Instruct* and InternVL *OpenGVLab/InternVL3_5-2B* received identical forced-choice classification and short, location-neutral caption instructions. YOLO-World *yolov8s-worldv2.pt* used the same six labels and a 0.25 threshold. Inputs followed the standard processors: 224 pixels for CLIP, 384 for BLIP, a 200,704-pixel Qwen budget, and 448-by-448 RGB for InternVL. Generation was deterministic; InternVL ran on an FAU A100 MIG and the other families in CPU environments.

Metrics and review. We report top-1 accuracy, frozen whole-term caption recall, and YOLO image-level hit rate; Dollar Street has no boxes, so mAP/IoU are inappropriate. Q4-minus-Q1 intervals use 2,000 category-stratified bootstrap resamples (seed 2026), and prompt contrasts pair images. Two raters independently reviewed all 336 BLIP captions while blinded to income and prompt, then adjudicated 49 captions with any disagreement. A separate one-rater 84-image Qwen audit was exploratory; InternVL had no semantic audit. All 1,344 model-task prediction rows passed coverage, uniqueness, and missing-value checks.

3. Results

Table 1: Primary scores (%); $n = 42$ images per quartile.

Model and task	Q1	Q2	Q3	Q4	All
CLIP classification	76.2	81.0	95.2	92.9	86.3
Qwen classification	85.7	88.1	83.3	88.1	86.3
InternVL classification	88.1	92.9	78.6	88.1	86.9
YOLO-World detection	11.9	14.3	14.3	21.4	15.5
BLIP caption, baseline	28.6	52.4	54.8	76.2	53.0
BLIP caption, prompted	26.2	54.8	45.2	78.6	51.2
Qwen caption	35.7	64.3	71.4	81.0	63.1
InternVL caption	42.9	59.5	66.7	64.3	58.3

Classifiers had similar overall accuracy but different income contrasts. CLIP’s Q4-minus-Q1 gap was 16.7 pp (CI 2.4 to 28.6); Qwen’s 2.4 pp and InternVL’s 0.0 pp intervals included zero. YOLO-World reached 15.5% overall with a 9.5 pp gap (-4.8 to 21.4). Automatic caption gaps were 47.6 pp for BLIP baseline, 52.4 pp for BLIP prompted, 45.2 pp for Qwen, and 21.4 pp for InternVL; all corresponding intervals excluded zero. CLIP’s exploratory maximum-minus-minimum gap was 19.0 pp, while its Q4-minus-Q1 correct-label-rank change was -0.26 (CI -0.62 to 0.07).

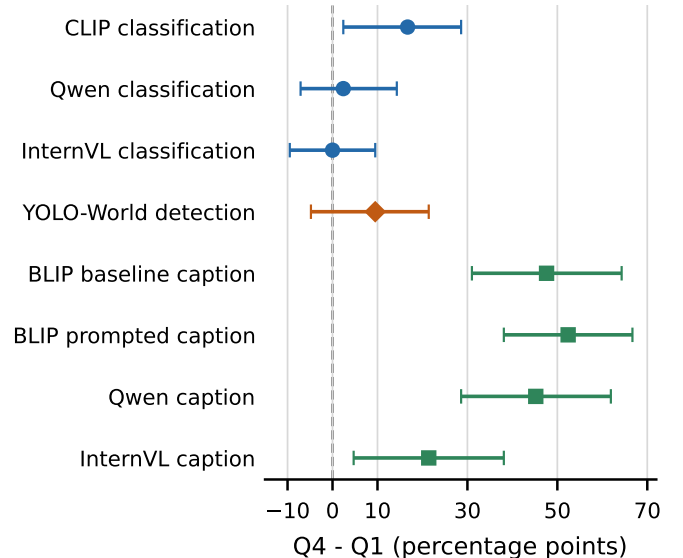


Figure 1: Q4-minus-Q1 gaps with 95% stratified bootstrap intervals; positive values favor Q4.

Prompt and human review. Prompting changed automatic BLIP recall by -1.8 pp overall (paired CI -7.1 to 3.6) and -2.4 pp in Q1 (-14.3 to 7.1). Human-rated correctness changed from 80.4% to 83.9% overall (+3.6 pp; -0.6 to 8.3), but from 78.6% to 73.8% in Q1 (-4.8 pp; -14.3 to 4.8). Hallucination changed from 16.7% to 12.5% overall (-4.2 pp; -8.9 to 0.6) and stayed 16.7% in Q1. Every prompt-effect interval included zero.

Across BLIP captions, consensus correctness was 82.1%, non-informativeness 0.9%, and hallucination 14.6%. Raw agreement/Cohen’s κ were 89.3%/0.646, 99.4%/0.498, and 94.0%/0.769, respectively. Combined correctness was 76.2% in Q1 and 89.3% in Q4. The exploratory Qwen audit found 70.2% clear semantic recall versus 57.1% automatic recall on its 84-image subset (11 uncertain cases), distinct from the 63.1% full-set value.

Failures. CLIP was weakest on light sources (60.7%); Qwen and InternVL classification were weakest on roofs (39.3% and 50.0%). YOLO-World found 57.1% of footwear and 25.0% of stoves, but no roofs, lights, or switches. Observed modes included unfamiliar local forms (a Q1 light source made from a red can was labelled stove), context dominance (a small roof was labelled light source), dictionary mismatch (valid words such as bucket, basket, and flip-flops were absent), and detections falling below the fixed threshold. Examples were retained across all quartiles rather than selected only from Q1.

4. Discussion and Limitations

There was no universal income-related pattern: CLIP and caption recall favored Q4, while Qwen and InternVL classification were nearly flat. Semantic review also showed that keyword recall understates caption correctness. The tested BLIP prefix was not reliably beneficial, particularly in Q1.

The contrast therefore depends on both the model and the evaluation rule. A caption can describe the right object with a synonym outside the dictionary, while a forced-choice classifier can succeed without producing a useful description. The positive caption gaps should be interpreted as disparities in this fixed term-recall measure, not as proof of lower general language quality for Q1 images.

The study is exploratory: 168 selected images, six classes, and seven observations per category-quartile cell. Balance does not control country, region, image quality, scene complexity, or within-class form; unequal regional scores are therefore descriptive only. Intervals are not multiplicity-adjusted. Income was never supplied to the models, so gaps are associations, not causal effects. The dictionary can miss valid wording; only BLIP had a complete blinded audit; and YOLO hit rate is not comparable to box-level benchmarks. Findings may change with other checkpoints, prompts, thresholds, or fine-tuning.

Demo and reproducibility. A local Gradio prototype compares uploaded images across the five model families. The locked manifest, exact prompts, package and device metadata, precomputed outputs, review files, failure examples, and figures are on the [project branch](#). The interface is demonstration evidence, not a hosted evaluation service or a second test set.

5. Conclusion

On this balanced benchmark, gaps depended on model, task, and metric. Human review confirmed that term recall understates semantic quality and that the BLIP prefix did not reliably improve the targeted Q1 outcome.

6. Qualitative Evidence

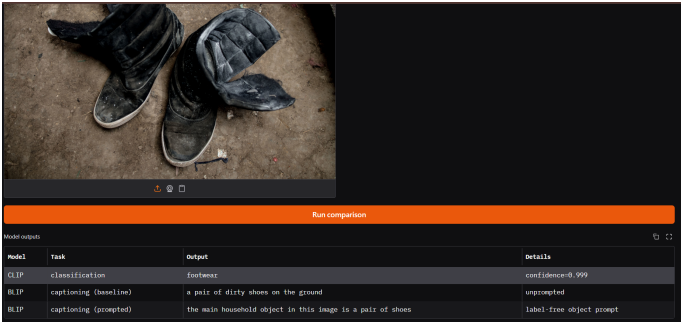


Figure 2: Local Gradio success: CLIP and both BLIP conditions recognize footwear.

In the success case, CLIP assigns the correct footwear label and both BLIP outputs describe shoes despite surface wear and a cluttered background. The failure case illustrates the complementary problem: the models describe visible appearance but miss the object function. These examples support the quantitative taxonomy without being used to estimate performance.

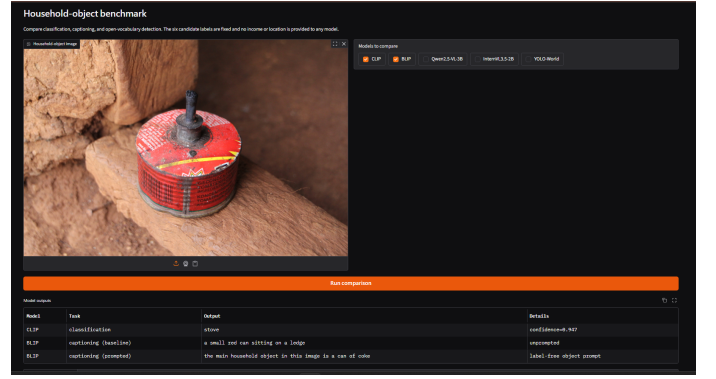


Figure 3: Failure example: a Q1 improvised light source is classified as a stove. Screenshots are qualitative evidence; Dollar Street images are CC BY 4.0.

7. Supplementary Checks

Table 2: Maximum-minus-minimum quartile gaps (percentage points).

Condition	Extrema	Gap	95% CI
CLIP class.	Q1–Q3	19.0	11.9, 31.0
Qwen class.	Q3–Q2	4.8	2.4, 16.7
InternVL class.	Q3–Q2	14.3	7.1, 26.2
YOLO detect.	Q1–Q4	9.5	4.7, 23.8
BLIP baseline	Q1–Q4	47.6	31.0, 64.3
BLIP prompted	Q1–Q4	52.4	38.1, 66.7
Qwen caption	Q1–Q4	45.2	28.6, 61.9
InternVL caption	Q1–Q3	23.8	11.9, 42.9

Maximum-minus-minimum intervals reuse the stratified bootstrap but are descriptive because extrema are selected.

Table 3: Scores (%) by region; columns show image counts.

Condition	AF 39	AM 44	AS 64	EU 21
CLIP class.	82.1	75.0	93.8	95.2
Qwen class.	79.5	86.4	89.1	90.5
InternVL class.	79.5	86.4	92.2	85.7
YOLO detect.	15.4	13.6	20.3	4.8
BLIP baseline	33.3	61.4	53.1	71.4
BLIP prompted	35.9	50.0	54.7	71.4
Qwen caption	43.6	70.5	60.9	90.5
InternVL caption	38.5	68.2	59.4	71.4

Regional values are uncontrolled because category and country composition differ; AF = Africa, AM = Americas, AS = Asia, and EU = Europe.

8. References

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- [4] J. Nwatu, O. Ignat, and R. Mihalcea. Bridging the Digital Divide: Performance Variation across Socio-Economic Factors in Vision-Language Models. *EMNLP*, 10686–10702, 2023.
- [5] A. Radford et al. Learning Transferable Visual Models from Natural Language Supervision. *ICML*, 8748–8763, 2021.
- [6] W. A. Gaviria Rojas et al. The Dollar Street Dataset: Images Representing the Geographic and Socioeconomic Diversity of the World. *NeurIPS Datasets and Benchmarks*, 2022.
- [7] W. Wang et al. InternVL3.5: Advancing Open-Source Multimodal Models. arXiv:2508.18265, 2025.

Appendix: Related Work

Datasets for socioeconomic visual diversity

Dollar Street pairs household photographs with country, region, and income metadata, making it possible to study visual variation that is absent from many web-image benchmarks [1]. Our work uses its public test split but constructs a locked, category-balanced subset: six household-object categories, four within-split income quartiles, and seven images in every category-quartile cell. This design controls category frequency while retaining the dataset's natural variation in country, scene, and object form.

Socioeconomic effects in vision-language models

Nwatu et al. [2] measured differences in CLIP image-text alignment and prompt-based retrieval across socioeconomic factors. We extend that question in three ways. First, we evaluate five model families rather than a single CLIP system. Second, we compare forced-choice classification, free-form captioning, and open-vocabulary detection on the same images. Third, we check whether automatic caption-term recall agrees with blinded human judgements, so a vocabulary mismatch is not treated as a model error without qualification.

Model families and task formulation

CLIP [3] provides the zero-shot contrastive baseline. BLIP [4] supplies image captioning and allows a fixed, training-free prompt comparison. Qwen2.5-VL [5] and InternVL3.5 [6] represent instruction-following multimodal models. YOLO-World [7] adds open-vocabulary detection. Unlike standard detection benchmarks, Dollar Street has no bounding-box annotations; we therefore report image-level correct-class detection and explicitly avoid mAP or IoU claims. The shared six-label setting lets us compare outputs while keeping prompts, images, and income grouping fixed.

Position of this study

The contribution is a small, reproducible benchmark rather than a claim about population-wide fairness. We report quartile gaps with category-stratified bootstrap intervals, retain negative and near-zero results, and separate term-based caption recall from human-rated semantic correctness. The resulting comparison shows that an apparent Q4 advantage for CLIP or caption recall does not automatically transfer to Qwen or InternVL classification, and that the measured conclusion depends on both task and evaluation rule.

References

- [1] W. A. Gaviria Rojas et al. “The Dollar Street Dataset: Images Representing the Geographic and Socioeconomic Diversity of the World.” NeurIPS Datasets and Benchmarks, 2022.
- [2] J. Nwatu, O. Ignat, and R. Mihalcea. “Bridging the Digital Divide: Performance Variation across Socio-Economic Factors in Vision-Language Models.” EMNLP, 2023.
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